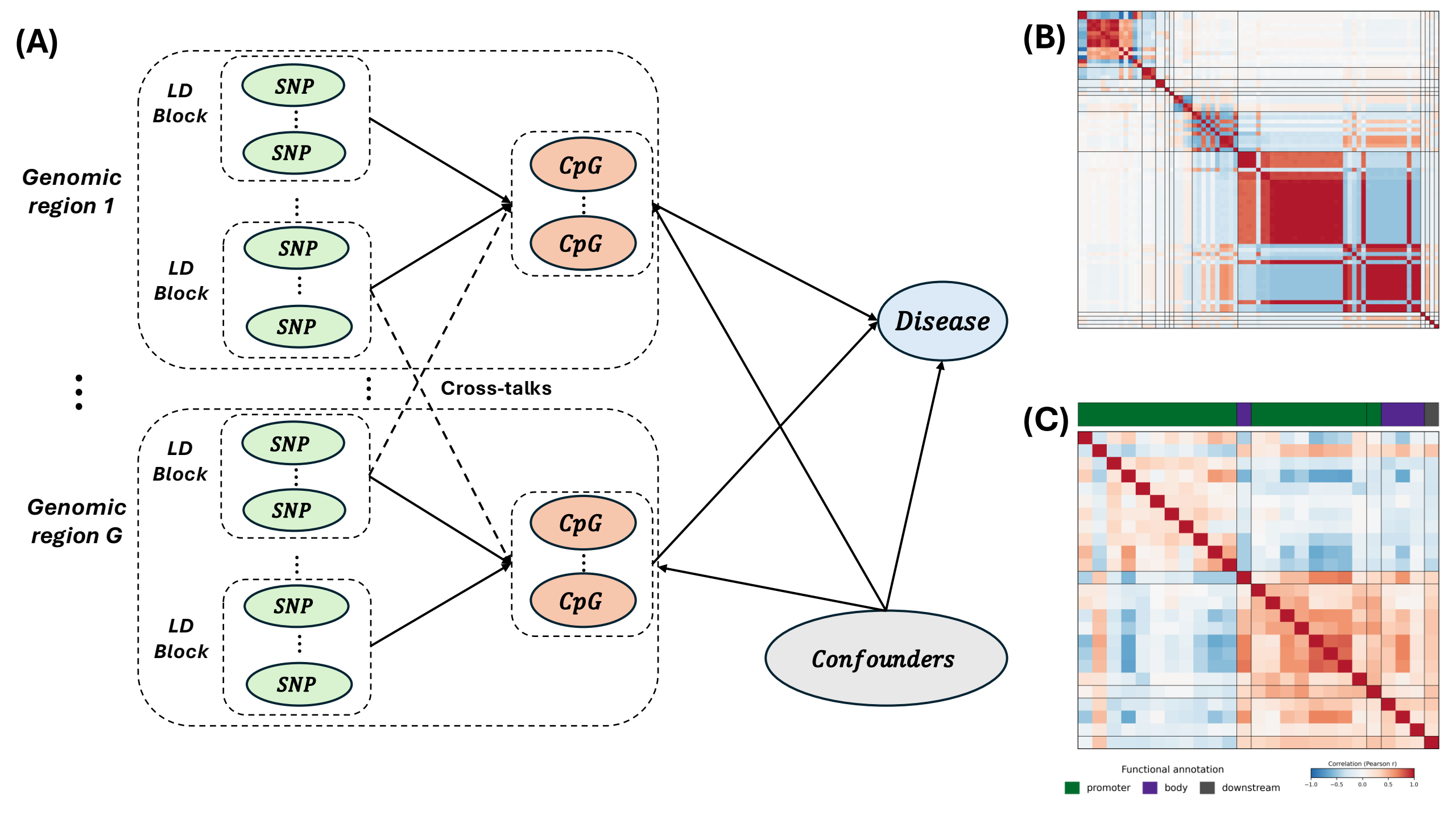
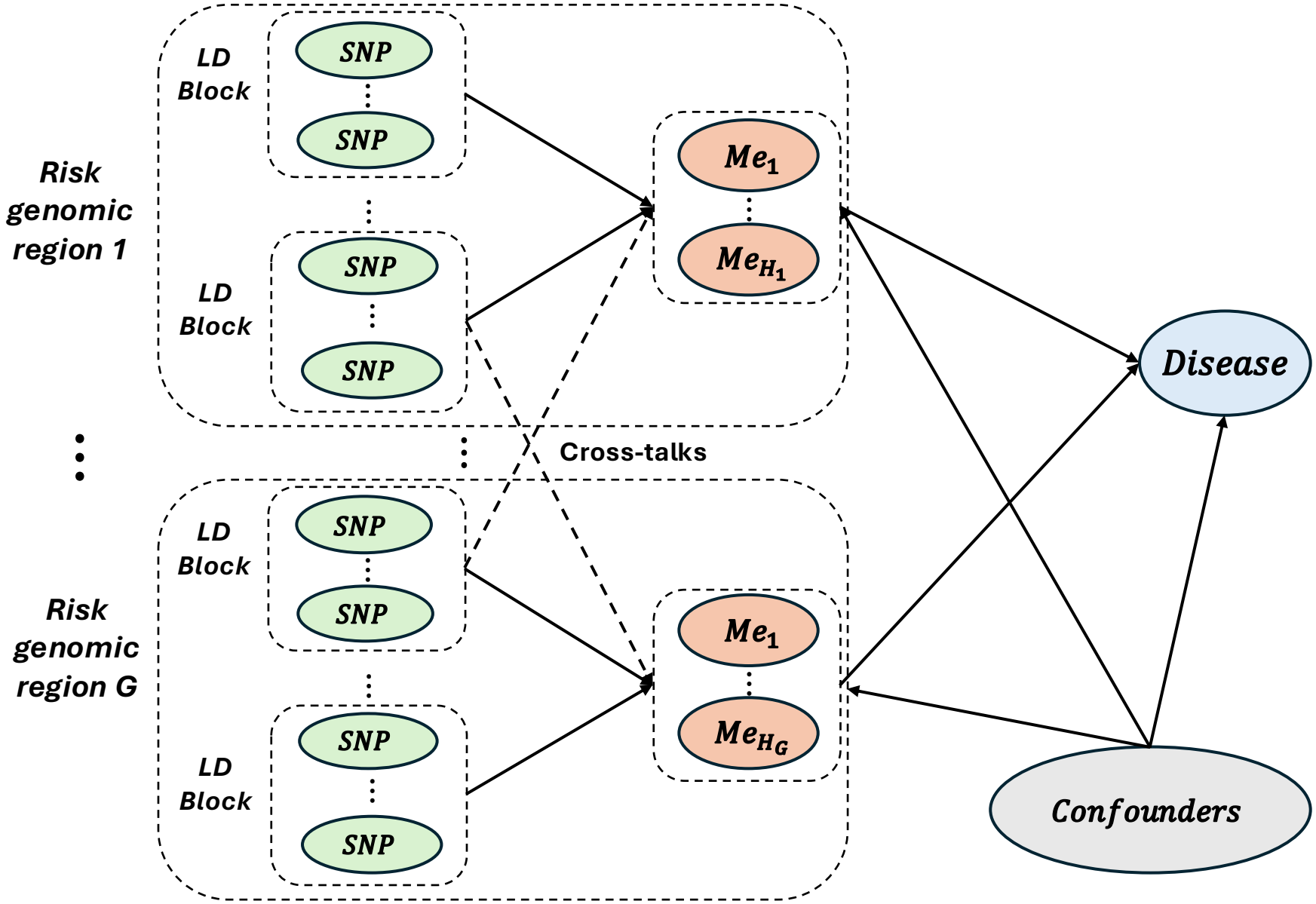
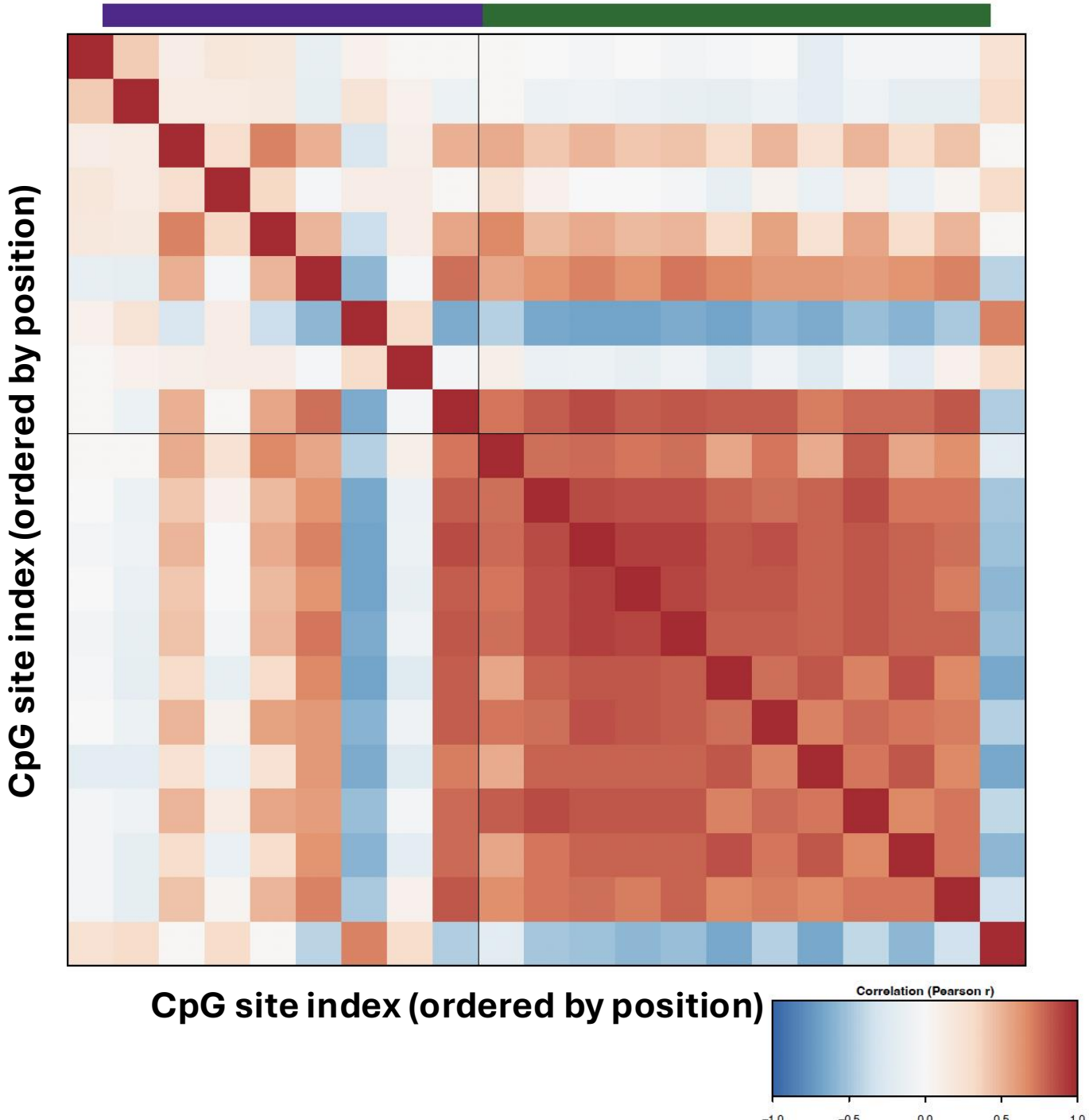
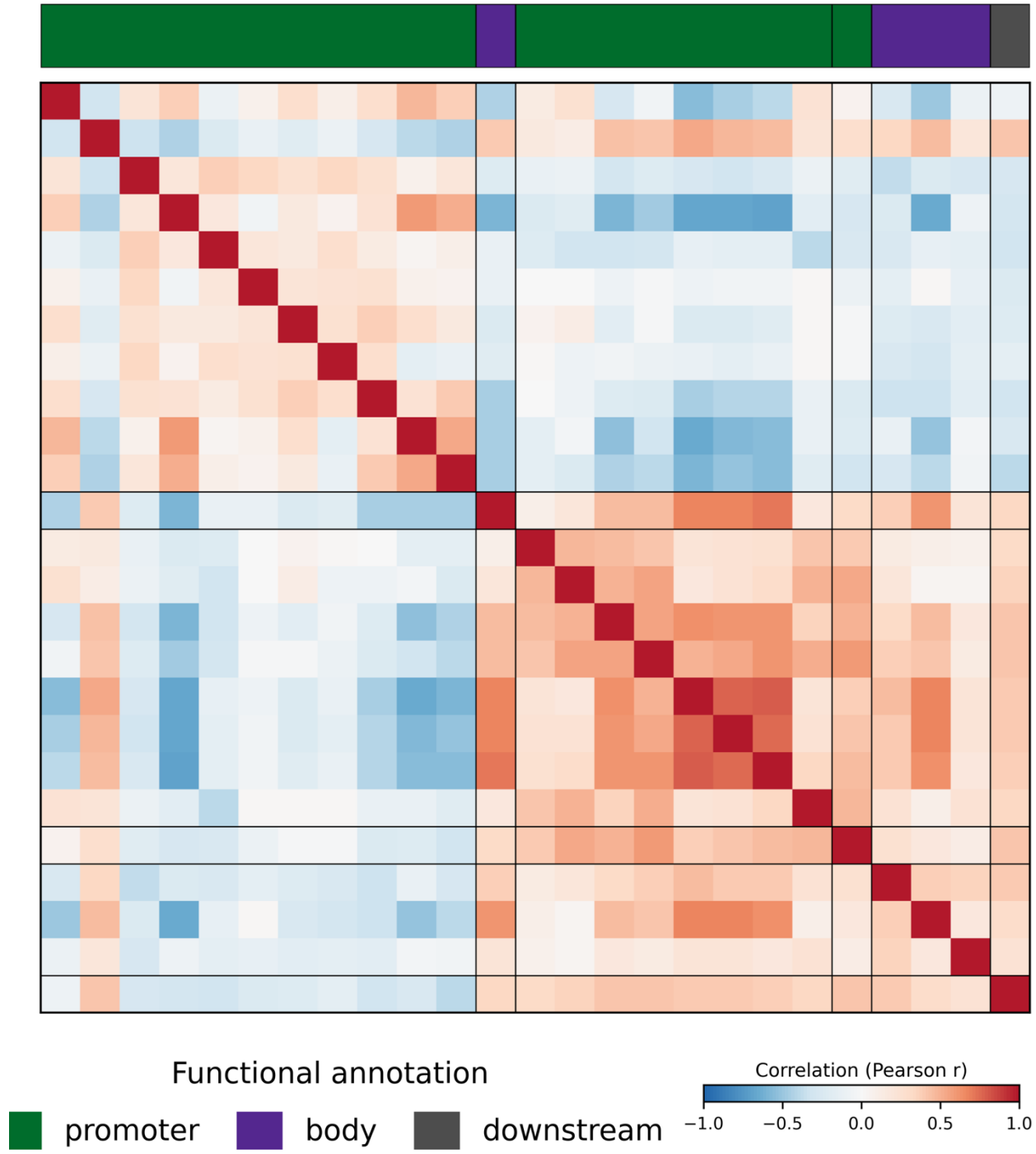
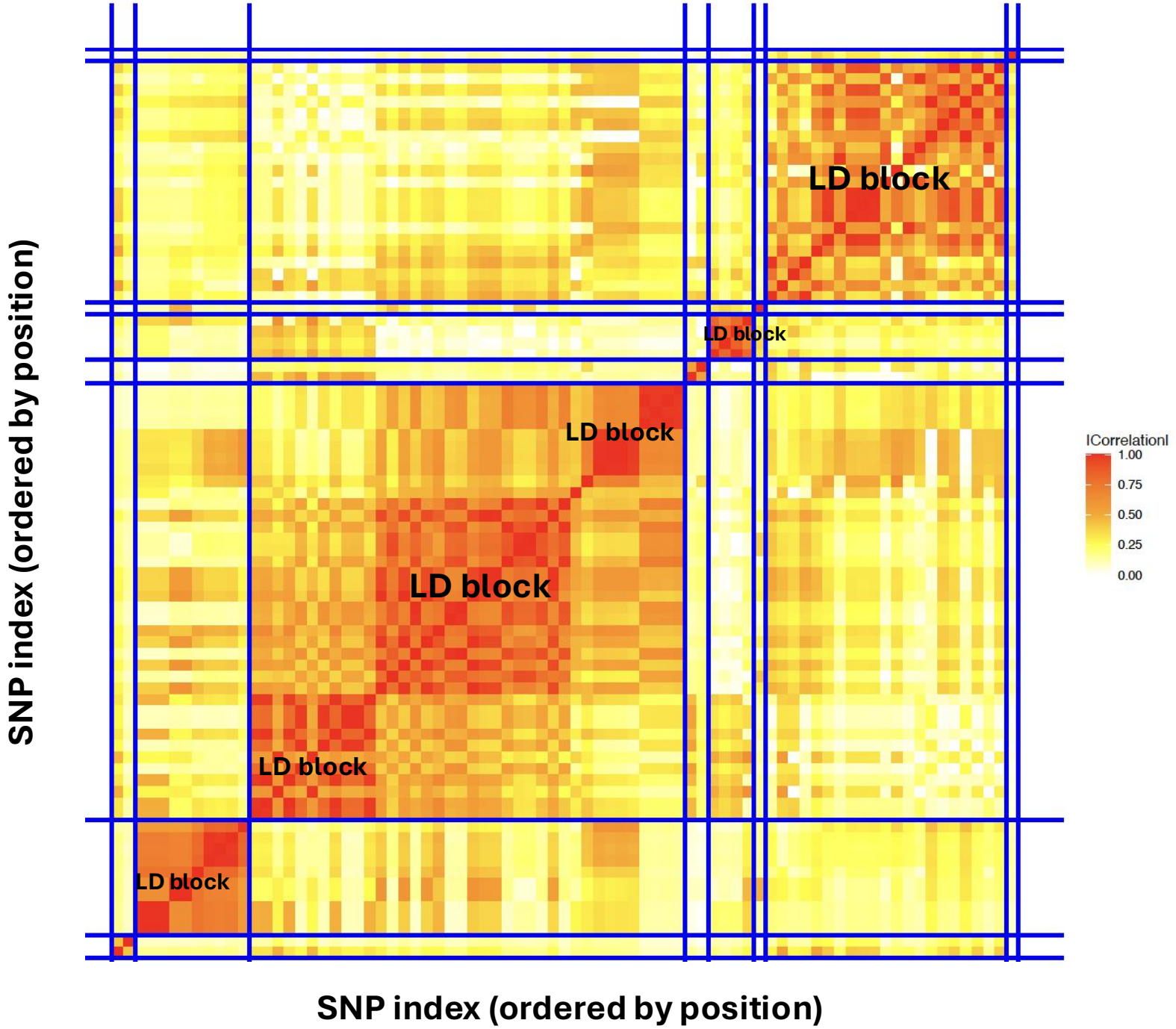
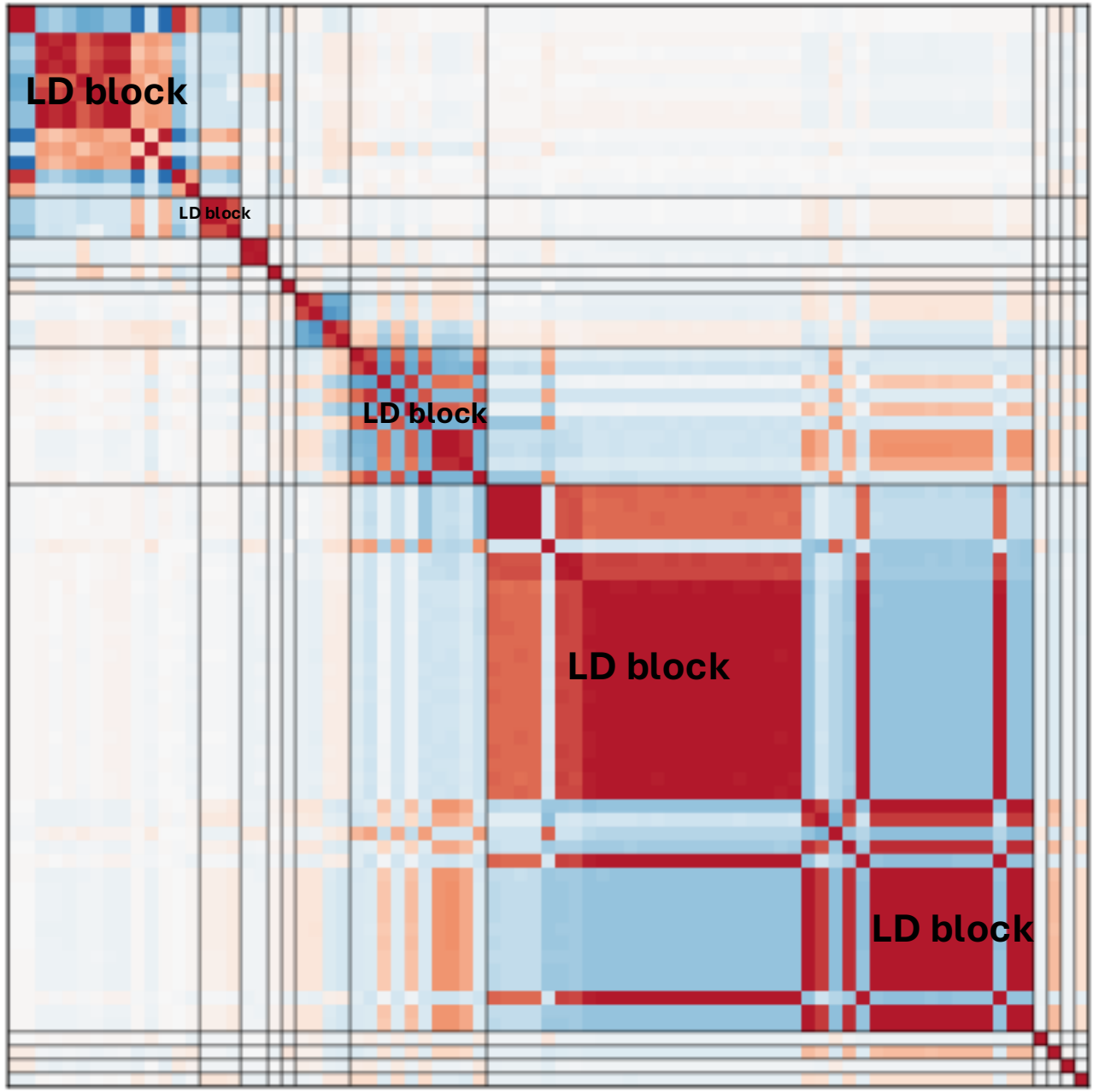

(A)
Genomic region 1
LD Block
SNP
SNP
LD Block
SNP
SNP
CpG
CpG
Cross-talks
Genomic region G
LD Block
SNP
SNP
LD Block
SNP
SNP
CpG
CpG
Disease
Confounders
(B)
(C)
Functional annotation
promoter
body
downstream
Correlation (Pearson r)
−1.0
−0.5
0.0
0.5
1.0

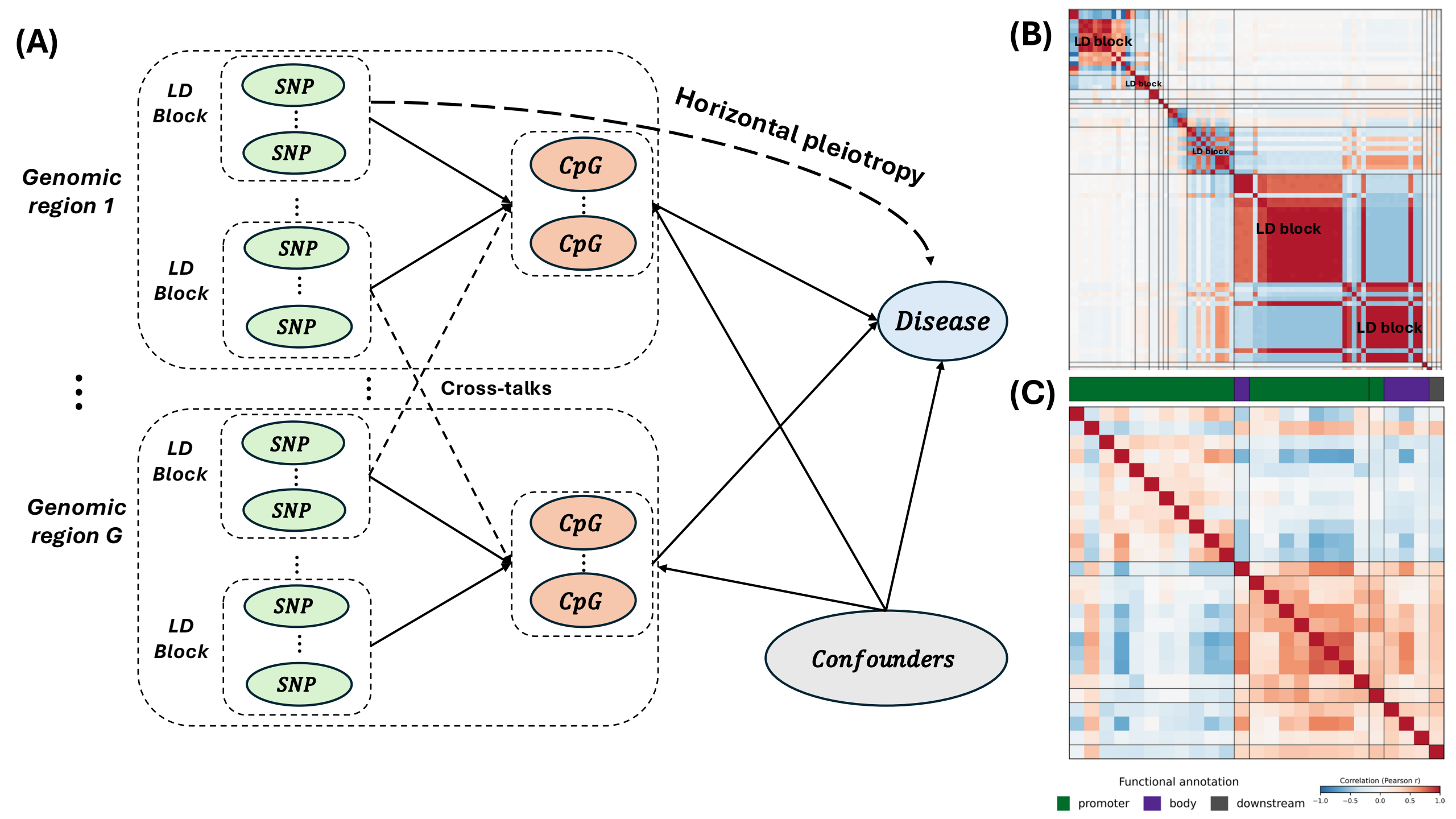

Risk genomic region 1
LD Block
SNP
SNP
LD Block
SNP
SNP
Me_1
Me_{H_1}
Cross-talks
Risk genomic region G
LD Block
SNP
SNP
LD Block
SNP
SNP
Me_1
Me_{H_G}
Disease
Confounders

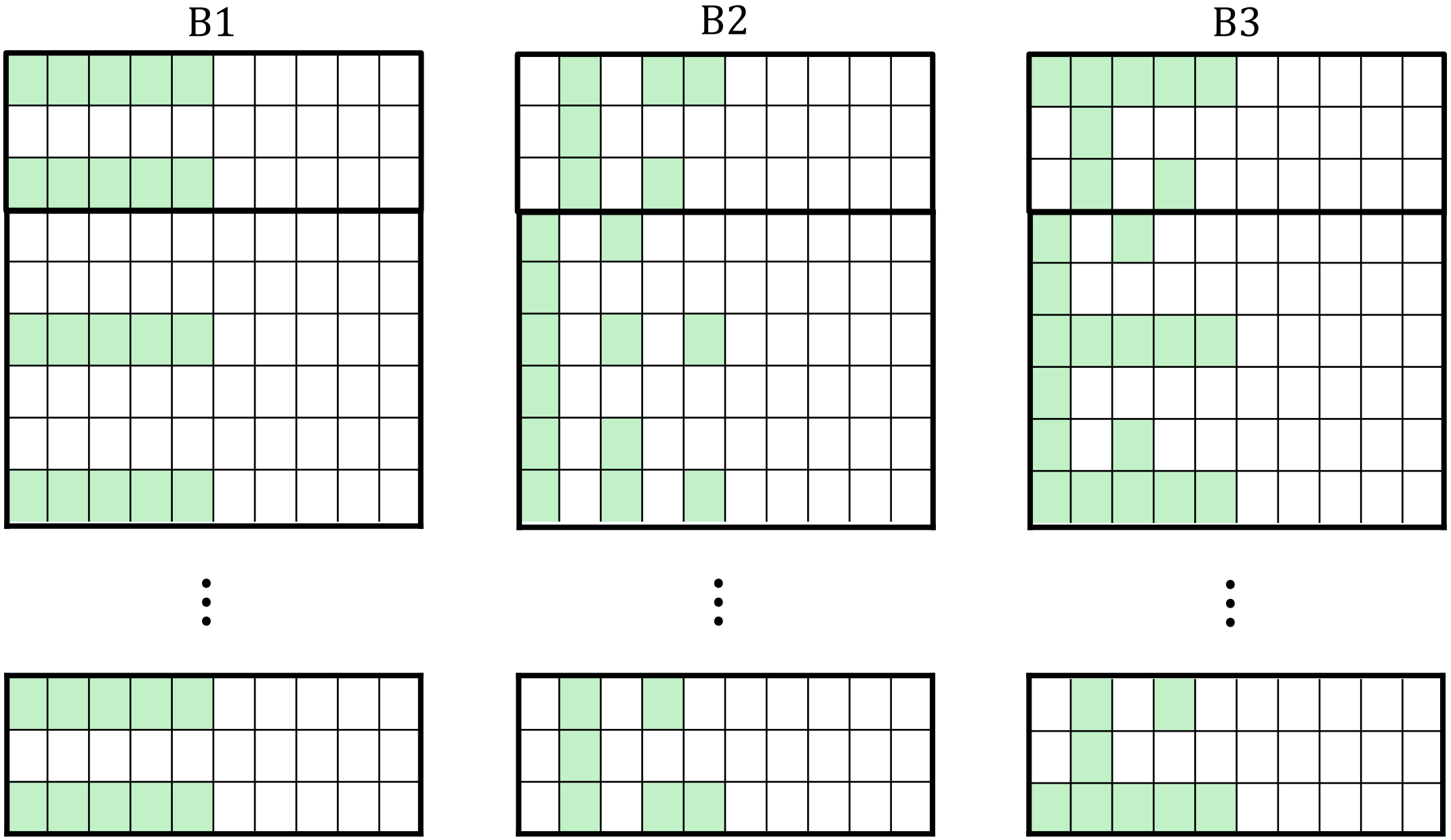

CpG site index (ordered by position)
CpG site index (ordered by position)
Correlation (Pearson r)

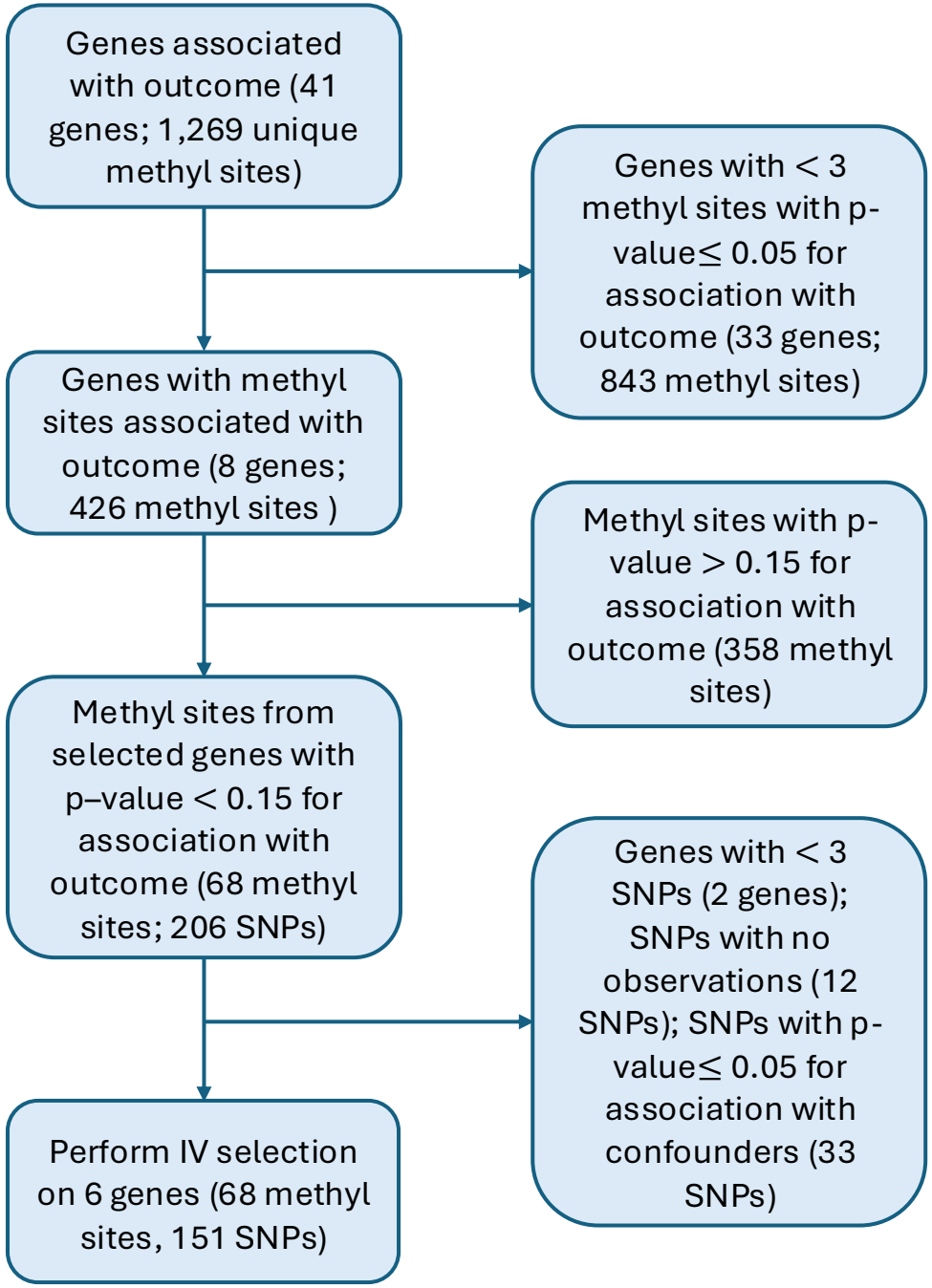

Functional annotation
promoter
body
downstream
Correlation (Pearson r)
−1.0
−0.5
0.0
0.5
1.0


LD block
LD block
LD block
LD block
LD block
LD block
|Correlation|
1.00
0.75
0.50
0.25
0.00
SNP index (ordered by position)
SNP index (ordered by position)


LD block
LD block
LD block
LD block
LD block


(A)
Genomic region 1
LD Block
SNP
SNP
LD Block
SNP
SNP
CpG
CpG
Horizontal pleiotropy
Disease
Cross-talks
Genomic region G
LD Block
SNP
SNP
LD Block
SNP
SNP
CpG
CpG
Confounders
(B)
LD block
LD block
LD block
LD block
LD block
(C)
Functional annotation
promoter
body
downstream
Correlation (Pearson r)
−1.0
−0.5
0.0
0.5
1.0


B1
B2
B3


Genes associated with outcome (41 genes; 1,269 unique methyl sites)
Genes with < 3 methyl sites with p-value≤ 0.05 for association with outcome (33 genes; 843 methyl sites)
Genes with methyl sites associated with outcome (8 genes; 426 methyl sites)
Methyl sites with p-value > 0.15 for association with outcome (358 methyl sites)
Methyl sites from selected genes with p–value < 0.15 for association with outcome (68 methyl sites; 206 SNPs)
Genes with < 3 SNPs (2 genes); SNPs with no observations (12 SNPs); SNPs with p-value≤ 0.05 for association with confounders (33 SNPs)
Perform IV selection on 6 genes (68 methyl sites, 151 SNPs)